

Curvature at Infinity of Scalar-Flat ALE Four-Manifolds

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- 1 Background
- 2 Main results
 - Preferred expansion
 - Tracking the Weyl part
 - Uniqueness
 - The Burns metric
- 3 Application to Kähler ALE surfaces

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ALE ends

An end $\Omega \subset (M^4, g)$ is ALE of order $\tau > 0$ if there are $R > 0$, a finite subgroup $\Gamma \subset \mathrm{SO}(4)$ acting freely on $\mathbb{R}^4 \setminus B_R$, and a diffeomorphism

$$\Phi : \Omega \rightarrow (\mathbb{R}^4 \setminus B_R)/\Gamma$$

such that, on the cover, $\tilde{g} = (\Phi^{-1} \circ \pi)^* g$ satisfies

$$\tilde{g}_{ij} = \delta_{ij} + \mathcal{O}(|x|^{-\tau}), \quad \partial^\alpha \tilde{g}_{ij} = \mathcal{O}(|x|^{-\tau-|\alpha|}).$$

The Tian–Viaclovsky baseline

Tian–Viaclovsky consider complete noncompact four-manifolds satisfying

$$\text{Scal}_g = 0, \quad \int_M |\text{Rm}_g|^2 dV_g < \infty,$$

together with finite Sobolev constant, $b_1(M) < \infty$, and one of the two curvature assumptions:

self-dual / anti-self-dual or harmonic curvature.

Theorem (Tian–Viaclovsky)

Each end is ALE, and in their exterior chart

$$g_{ij} = \delta_{ij} + \mathcal{O}_\infty(r^{-\tau}) \quad \text{for every } \tau < 2.$$

Where scalar-flat Kähler ALE surfaces enter

Kähler ALE surfaces

In real dimension four, a scalar-flat Kähler metric is anti-self-dual with respect to the Kähler orientation.

Hence scalar-flat Kähler ALE surfaces are included in the self-dual / anti-self-dual branch of the Tian–Viaclovsky theorem.

Starting point

Tian–Viaclovsky gives

$$g_{ij} = \delta_{ij} + \mathcal{O}_\infty(r^{-\tau}) \quad \text{for every } \tau < 2.$$

The decay is almost r^{-2} , but the leading coefficient is still not determined.

Known results

Here are the known results most closely related to this talk:

- In 1989, Bando–Kasue–Nakajima constructed coordinates at infinity under fast curvature decay and maximal volume growth.
- In 1994, Cheeger–Tian proved the cone structure at infinity for Ricci-flat manifolds with Euclidean volume growth and quadratic curvature decay.
- In 2005–2008, Tian–Viaclovsky proved ALE structure and curvature decay for four-dimensional critical metrics, including the scalar-flat classes considered here.
- In 2023, Biquard–Hein related the leading asymptotics of Ricci-flat ALE spaces to renormalized volume.
- In 2025, Wang–Yin constructed preferred coordinates and a Weyl tensor at infinity for Ricci-flat ALE ends.
- In Kähler ALE geometry, Kronheimer, Calderbank–Singer, Lock–Viaclovsky, Han–Viaclovsky, Hein–Räsdeaconu–Şuvaina, and Hein–LeBrun provide the main examples, moduli results, and mass formula.

Wang–Yin: the key idea

Wang–Yin study preferred coordinates for Ricci-flat ends at infinity.

The technical point we use as a model is the following strategy:

- ① put the metric in a suitable Bianchi-type gauge at infinity;
- ② extract the leading $|x|^{-m}$ coefficient from the elliptic equation;
- ③ use decaying coordinate changes

$$x = y + u(y), \quad u_i(y) = B_{ij} \frac{y_j}{|y|^m};$$

- ④ separate the coordinate freedom from the geometric part of the leading coefficient.

Point for this talk

We adapt this gauge-and-coordinate strategy to the scalar-flat ALE setting.

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Notation for the main theorem

For $\lambda \in \mathbb{R}$, define the scalar pair-symmetric tensor

$$\Xi(\lambda)_{ijkl} = -\frac{\lambda}{9}\delta_{ij}\delta_{kl} + \frac{2\lambda}{9}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}).$$

Fix an ALE end modeled on $(\mathbb{R}^4 \setminus B_R)/\Gamma$. Set

$$m_{\text{Sch}} = |\Gamma|m_{\text{ADM}}(g), \quad U(x) = 1 + \frac{m_{\text{Sch}}}{2|x|^2}, \quad g^{\text{Sch}} = U^2 g_{\text{E}}.$$

The normalized Schwarzschild reference metric is

$$g^{\text{Sch},\sharp} = (\Phi^{\text{Sch}})^* g^{\text{Sch}}, \quad \Phi^{\text{Sch}}(x) = x - m_{\text{Sch}} \frac{x}{|x|^2}.$$

Main theorem: preferred coordinates

Theorem (Preferred expansion)

Let (M^4, g) satisfy the Tian–Viaclovsky assumptions, and fix one ALE end modeled on $(\mathbb{R}^4 \setminus B_R)/\Gamma$. Then there exist $\sigma \in (0, 1)$ and preferred coordinates x on the lifted end such that

$$g_{ij}(x) = \delta_{ij} + (\Xi(9|\Gamma|m_{\text{ADM}}(g))_{ijkl} + (W_\infty)_{ik\ell j}) \frac{x_k x_\ell}{|x|^4} + \mathcal{O}_\infty(|x|^{-2-\sigma}),$$

where W_∞ is an algebraic Weyl tensor.

The same coordinates satisfy the relative Schwarzschild gauge

$$g^{ij} \left(\Gamma(g)_{ij}^k - \Gamma(g^{\text{Sch}, \#})_{ij}^k \right) = 0.$$

Proof strategy

The proof has two stages.

Analytic stage

Use Euclidean Bianchi gauge. The scalar-flat self-dual / anti-self-dual equation and the scalar-flat harmonic-curvature equation both become

$$\frac{1}{4}\Delta_{g_E}^2 h = N(h).$$

This gives the rough expansion

$$h_{ij} = A_{ijkl} \frac{x_k x_\ell}{r^4} + \mathcal{O}_\infty(r^{-3}).$$

Algebraic stage

Use the characteristic properties of the Weyl-type tensor to isolate it from the coefficient A , a process that effectively removes the coordinate freedom inherent in A .

Analytic reduction at infinity

Start from the Tian–Viaclovsky ALE chart

$$g = g_E + h_0, \quad h_0 = \mathcal{O}_\infty(r^{-\tau}), \quad \tau < 2.$$

After a small coordinate change, we put the perturbation $h = g - g_E$ in the Euclidean Bianchi gauge:

$$B_{g_E}(h) := \operatorname{div}_{g_E} h - \frac{1}{2}d(\operatorname{tr}_{g_E} h) = 0.$$

In this gauge, both curvature assumptions in the Tian–Viaclovsky setting have the same leading elliptic form:

$$\frac{1}{4}\Delta_{g_E}^2 h = N(h),$$

where $N(h)$ consists of lower-order nonlinear terms.

The first coefficient

The elliptic reduction gives a rough leading expansion:

Proposition (Rough leading term)

There exist $R \gg 1$ and a constant tensor $A = (A_{ijkl})$ such that

$$h_{ij}(x) = A_{ijkl} \frac{x_k x_\ell}{r^4} + \mathcal{O}_\infty(r^{-3}) \quad \text{on } \Omega_R.$$

The Bianchi gauge already imposes restrictions on A :

$$A_{ijkl} = A_{jikl} = A_{ijlk}, \quad 2A_{ijj\ell} - 2A_{ilkk} - A_{ppil} + \delta_{il}A_{ppkk} = 0.$$

Next step

The tensor A is constrained, but it still contains coordinate freedom. The rest of the proof separates this coordinate freedom from the geometric part of the leading coefficient.

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The algebraic target

Let $\mathcal{W}$ be the space of algebraic Weyl tensors on $\mathbb{R}^4$. Define

$$s : \mathcal{C} \longrightarrow S^2(\mathbb{R}^4) \otimes S^2(\mathbb{R}^4), \quad s(R)_{ijkl} = \frac{1}{2}(R_{iklj} + R_{iljk}).$$

Write $\widetilde{\mathcal{W}} = s(\mathcal{W})$.

Proposition (Weyl-type criterion)

A tensor $A \in S^2(\mathbb{R}^4) \otimes S^2(\mathbb{R}^4)$ lies in $\widetilde{\mathcal{W}}$ if and only if

- ① $A_{ijkl} = A_{jikl} = A_{ijlk}$;
- ② $A_{ppkl} = 0$ and $A_{ijjl} = 0$;
- ③ $A_{ijkl} + A_{iklj} + A_{iljk} = 0$.

These are the three tests we impose on the rough coefficient.

Recovering the Weyl tensor

If the three identities hold, then

$$A = s(W_\infty)$$

for a unique algebraic Weyl tensor W_∞ .

It is recovered by an explicit inverse formula:

$$W_\infty = -\frac{1}{3}\mathbf{R}A,$$

where

$$(\mathbf{R}A)_{ijkl} = A_{jlik} - A_{jkil} - A_{iljk} + A_{ikjl}.$$

Consequence

Once the coefficient A satisfies the three algebraic tests, the Weyl tensor at infinity is forced.

The idea

Use the Weyl-type criterion as a checklist:

$$\text{pair symmetry} \implies \text{partial trace-free} \implies \text{first Bianchi.}$$

The first condition is already true for the rough coefficient A .

The essential problem is the second condition:

$$A_{ppk\ell} = 0, \quad A_{ijj\ell} = 0.$$

The coordinate change

Use the similar decaying coordinate change as in Wang–Yin:

$$x = \varphi(y) = y + u(y), \quad u_i(y) = B_{ij} \frac{y_j}{|y|^2}.$$

At the level of the $|x|^{-2}$ coefficient,

$$A \longmapsto A + \Psi_4(B),$$

where

$$\begin{aligned} \Psi_4(B)_{ijkl} = & (B_{ij} + B_{ji})\delta_{kl} \\ & - (B_{ik}\delta_{jl} + B_{il}\delta_{jk} + B_{jk}\delta_{il} + B_{jl}\delta_{ik}). \end{aligned}$$

Use

Choose B to eliminate as much trace as possible.

The obstruction

Let

$$U_{k\ell} = A_{ppk\ell}, \quad V_{i\ell} = A_{ijj\ell}.$$

We would like to use a decaying coordinate change to make

$$(A + \Psi_4(B))_{ppk\ell} = 0, \quad (A + \Psi_4(B))_{ijj\ell} = 0.$$

This system is solvable only under two conditions:

$$(V_{(i\ell)} - U_{i\ell})_0 = 0, \quad \text{tr } V = -2 \text{tr } U.$$

Key point

Scalar-flatness gives the first condition, but not the trace condition. Thus a scalar obstruction remains:

$$\lambda(A) = \frac{1}{4}(\text{tr } V + 2 \text{tr } U).$$

Relaxing the target

Since full partial trace-free elimination is obstructed, we weaken the target.

Instead of requiring both partial traces to vanish, we seek coordinates such that

$$A_{ppk\ell}^\# = 0, \quad A_{ijj\ell}^\# = A_{\ell jj i}^\#, \quad A^\# = A + \Psi_4(B).$$

Idea

Kill the first partial trace, symmetrize the second one, and keep track of the remaining scalar trace separately.

This weaker target is solvable by choosing

$$\mathrm{tr} B = -\frac{1}{4} \mathrm{tr} U, \quad B_{(k\ell)} = \frac{1}{4} U_{k\ell} - \frac{1}{8} \mathrm{tr} U \delta_{k\ell}, \quad B_{[i\ell]} = \frac{1}{6} V_{[i\ell]}.$$

Scalar part and Weyl-type remainder

Define

$$W(A) := A^\sharp - \Xi(\lambda(A)),$$

where

$$\Xi(\mu)_{ijkl} = -\frac{\mu}{9}\delta_{ij}\delta_{kl} + \frac{2\mu}{9}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}).$$

The scalar tensor is chosen so that

$$\Xi(\mu)_{ppkl} = 0, \quad \Xi(\mu)_{ijjl} = \mu\delta_{il}.$$

Thus subtracting $\Xi(\lambda(A))$ removes the remaining scalar trace:

$$W(A)_{ppkl} = 0, \quad W(A)_{ijjl} = 0.$$

The Bianchi-gauge identity further gives

$$W(A)_{ijkl} + W(A)_{iklj} + W(A)_{iljk} = 0.$$

Conclusion

$W(A)$ is Weyl-type. Hence there is a unique algebraic Weyl tensor W_∞ such that $W(A) = s(W_\infty)$.

Identification of the scalar part

At this point

$$A^\sharp = \Xi(\lambda(A)) + s(W_\infty).$$

The remaining scalar is computed from the ALE ADM mass:

$$m_{\text{ADM}}(g) = \frac{1}{6\omega_3|\Gamma|} \lim_{r \rightarrow \infty} \int_{S_r} (\partial_j g_{ij} - \partial_i g_{jj}) \nu_i \, d\mu.$$

For the homogeneous coefficient $A^\sharp$ one obtains

$$\lambda(A) = 9|\Gamma|m_{\text{ADM}}(g).$$

Canonical leading coefficient

$$A^\sharp = \Xi(9|\Gamma|m_{\text{ADM}}(g)) + s(W_\infty).$$

Why a Schwarzschild reference is needed

The coordinate transformation producing $A^\sharp$ does not preserve the Euclidean Bianchi gauge.

A direct Euclidean repair would require a vector field of order $|x|^{-1}$, hence it would change the $|x|^{-2}$ coefficient.

So the scalar part is absorbed into a Schwarzschild reference metric:

$$m_{\text{Sch}} = |\Gamma| m_{\text{ADM}}(g), \quad U(x) = 1 + \frac{m_{\text{Sch}}}{2r^2},$$

$$g^{\text{Sch}} = U^2 g_{\text{E}}, \quad \Phi_{\text{Sch}}(x) = x - m_{\text{Sch}} \frac{x}{r^2},$$

$$g^{\text{Sch}, \sharp} := \Phi_{\text{Sch}}^* g^{\text{Sch}}.$$

Then impose

$$g^{ij} (\Gamma(g)_{ij}^k - \Gamma(g^{\text{Sch}, \sharp})_{ij}^k) = 0.$$

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Uniqueness of W_∞

Theorem (Uniqueness)

Let x and y be two **preferred** coordinate systems on the same lifted ALE end. Suppose

$$g_{ij}^{(x)} = \delta_{ij} + \left(\Xi(9|\Gamma|m_{\text{ADM}})_{ijkl} + (W_\infty^{(x)})_{iklj} \right) \frac{x_k x_\ell}{|x|^4} + \mathcal{O}_\infty(|x|^{-2-\sigma}),$$

$$g_{ab}^{(y)} = \delta_{ab} + \left(\Xi(9|\Gamma|m_{\text{ADM}})_{abpq} + (W_\infty^{(y)})_{apqb} \right) \frac{y_p y_q}{|y|^4} + \mathcal{O}_\infty(|y|^{-2-\sigma}).$$

Then there is $A \in O(4)$ such that

$$(W_\infty^{(x)})_{ijkl} = A_{ai} A_{bj} A_{ck} A_{d\ell} (W_\infty^{(y)})_{abcd}.$$

Idea of the uniqueness proof

Let

$$F := y \circ x^{-1}.$$

Because F is the transition map between two expressions of the same metric, its components solve a harmonic-map type equation.

First estimates give

$$F(x) = Ax + b + \mathcal{O}_\infty(|x|^{-1}), \quad A \in O(4).$$

The relative Schwarzschild gauge improves the Euclidean Poisson equation to

$$\Delta_{g_E} F = \mathcal{O}_\infty(|x|^{-4}).$$

A Poisson estimate on the exterior region then gives

$$F(x) = Ax + b + \mathcal{O}_\infty(|x|^{-2} \log |x|).$$

Then

Comparing the two $|x|^{-2}$ coefficients gives the $O(4)$ transformation law for W_∞ .

Meaning of the theorem

The mass coefficient is fixed:

$$9|\Gamma|m_{\text{ADM}}(g).$$

The Weyl tensor at infinity is fixed modulo the natural orthogonal action:

$$W_\infty \in \mathcal{W}/\text{O}(4).$$

Interpretation

The theorem defines a curvature tensor at infinity for scalar-flat ALE ends in the Tian–Viaclovsky class.

quantity	behavior under preferred coordinate change
m_{ADM}	invariant
W_∞	transforms by the $\text{O}(4)$ action

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Example: the Burns metric

The Burns metric on $\text{Bl}_0 \mathbb{C}^2$ is scalar-flat Kähler ALE with $\Gamma = \{1\}$. In the standard ALE coordinates,

$$g_{ij}^{\text{Burns}} = \delta_{ij} + \frac{1}{r^2} \delta_{ij} - \frac{x_i x_j}{r^4} - \frac{(Jx)_i (Jx)_j}{r^4}.$$

This coefficient is very close to satisfying the Bianchi-gauge condition. We then use the coordinate transformation

$$x_i = y_i - \frac{1}{2} \frac{y_i}{|y|^2}.$$

Then

$$(A^{\text{Burns}})^{\sharp} = \Xi(3) + s(W_{\infty}^{\text{Burns}}), \quad W_{\infty}^{\text{Burns}} \neq 0.$$

What the Burns metric shows

The Burns metric has both parts of the preferred coefficient:

mass part $\neq 0$, Weyl-type part $\neq 0$.

Test case for the main theorem

It verifies that the preferred expansion is not merely a Weyl normal form. The scalar mass part is genuinely necessary in the scalar-flat ALE setting.

Test case for uniqueness

A further preferred coordinate change can only rotate $W_{\infty}^{\text{Burns}}$ by $O(4)$; it cannot remove it.

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Motivation

Lock–Viaclovsky construct scalar-flat Kähler ALE metrics on minimal resolutions of quotient surface singularities.

Let

$$\Gamma \subset U(2), \quad Y = \mathbb{C}^2/\Gamma, \quad \pi : X \rightarrow Y$$

be the minimal resolution.

Question

Can the Weyl tensor at infinity detect algebraic information about the resolution?

Answer

Yes. For minimal quotient resolutions,

$$W_\infty = 0 \iff K_X = \pi^* K_Y.$$

The Kähler coefficient at infinity

Use holomorphic ALE coordinates induced by the resolution map. The Kähler potential has the form

$$\omega = \sqrt{-1} \partial \bar{\partial} (\rho + c \log \rho + \psi), \quad \rho = |z|^2, \quad \psi = \mathcal{O}_\infty(r^{-1}).$$

Hence, in real coordinates u ,

$$g_{ij} = \delta_{ij} + c \left(\frac{\delta_{ij}}{|u|^2} - \frac{u_i u_j}{|u|^4} - \frac{(J_\infty u)_i (J_\infty u)_j}{|u|^4} \right) + \mathcal{O}_\infty(|u|^{-3}).$$

The logarithmic coefficient is

$$c = 3|\Gamma| m_{\text{ADM}}(g).$$

Kähler Weyl direction

Applying our coefficient-tracking procedure gives

$$(A^K)^\sharp = \Xi(9|\Gamma|m_{\text{ADM}}(g)) + s(3|\Gamma|m_{\text{ADM}}(g)W_{J_\infty}).$$

Therefore

$$W_\infty^{(x)}(g) = 3|\Gamma|m_{\text{ADM}}(g)W_{J_\infty^{(x)}}.$$

The tensor W_J is determined by the limiting complex structure:

$$\begin{aligned}(W_J)_{ijkl} &= \frac{1}{3}(\delta_{il}\delta_{jk} - \delta_{ik}\delta_{jl}) + \frac{2}{3}J_{ij}J_{kl} \\ &\quad + \frac{1}{3}J_{ik}J_{jl} - \frac{1}{3}J_{il}J_{jk}.\end{aligned}$$

Hein–LeBrun mass formula

For scalar-flat Kähler ALE surfaces,

$$m_{\text{ADM}}(g) = -\frac{1}{3\pi} \langle \clubsuit(c_1(X, J)), [\omega] \rangle.$$

Combining this with the previous formula gives

$$W_{\infty}^{(x)}(g) = -\frac{|\Gamma|}{\pi} \langle \clubsuit(c_1(X, J)), [\omega] \rangle W_{J_{\infty}^{(x)}}.$$

Reduction

Vanishing of W_{∞} is equivalent to vanishing of a Chern–Kähler pairing.

Discrepancy data of the resolution

Write the exceptional divisor as

$$E = \bigcup_{i=1}^k E_i, \quad E_i^2 = -e_i.$$

Let $Q = (E_i \cdot E_j)$ and $H = -Q$.

Discrepancy vector

$$K_X - \pi^* K_Y = - \sum_{i=1}^k \mu_i E_i, \quad H\mu = (e_1 - 2, \dots, e_k - 2)^T.$$

Kähler areas and pairing

$$A_i = \int_{E_i} \omega > 0, \quad \langle \clubsuit(c_1(X, J)), [\omega] \rangle = \sum_{i=1}^k \mu_i A_i.$$

Main application: detecting crepancy

Theorem (Weyl tensor at infinity detects crepancy)

Let $\Gamma \subset U(2)$ be finite and act freely on $\mathbb{S}^3$. Set $Y = \mathbb{C}^2/\Gamma$, and let $\pi : X \rightarrow Y$ be a minimal resolution. Let g be a scalar-flat Kähler ALE metric on (X, J) whose end is determined by π . For any preferred coordinate system x obtained from the holomorphic ALE coordinates, the following are equivalent:

- ① $W_{\infty}^{(x)}(g) = 0$ for one, equivalently for every, such x ;
- ② $\pi : X \rightarrow Y$ is crepant, equivalently $K_X = \pi^* K_Y$.

Why positivity proves it

The formula is

$$W_{\infty}^{(x)}(g) = -\frac{|\Gamma|}{\pi} \left(\sum_i \mu_i A_i \right) W_{J_{\infty}^{(x)}}.$$

For a minimal resolution, $e_i \geq 2$. Thus

$$(e_i - 2)_i \geq 0.$$

Since $H = -Q$ is an irreducible symmetric nonsingular M -matrix,

$$(H^{-1})_{ij} > 0.$$

Therefore, if the resolution is not crepant, then

$$\mu_i > 0 \quad \text{for all } i, \quad \sum_i \mu_i A_i > 0.$$

Hence W_{∞} cannot vanish.

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Thank you!